

SCR2043 OPERATING SYSTEMS

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Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. Download necessary source codes:

Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. Compile the source files:

Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

3. Execute the dummy processes:

Run all the dummy processes

```
./mainprocess &
```

Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

Lab Assessment 2 : Linux Process Monitoring and Management

Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

Question 1

Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

Command
<code>ps -e</code>
Output

```

syahmi@secl2043:~$ ps -e
  PID TTY          TIME CMD
    1 ?           00:00:01 systemd
    2 ?           00:00:00 kthreadd
    3 ?           00:00:00 pool_workqueue_release
    4 ?           00:00:00 kworker/R-rcu_g
    5 ?           00:00:00 kworker/R-rcu_p
    6 ?           00:00:00 kworker/R-slub_
    7 ?           00:00:00 kworker/R-netns
    8 ?           00:00:02 kworker/0:0-events
   11 ?           00:00:00 kworker/u4:0-floppy
   12 ?           00:00:00 kworker/R-mm_pe
   13 ?           00:00:00 rcu_tasks_kthread
   14 ?           00:00:00 rcu_tasks_rude_kthread
   15 ?           00:00:00 rcu_tasks_trace_kthread
   16 ?           00:00:00 ksoftirqd/0
   17 ?           00:00:00 rcu_preempt
   18 ?           00:00:00 migration/0
   19 ?           00:00:00 idle_inject/0
   20 ?           00:00:00 cpuhp/0
   21 ?           00:00:00 cpuhp/1
   22 ?           00:00:00 idle_inject/1
   23 ?           00:00:00 migration/1
   24 ?           00:00:00 ksoftirqd/1
   25 ?           00:00:03 kworker/1:0-events
   26 ?           00:00:00 kworker/1:0H-events_highpri
   29 ?           00:00:00 kdevtmpfs
   30 ?           00:00:00 kworker/R-inet_
   31 ?           00:00:00 kworker/u5:1-flush-8:0
   32 ?           00:00:00 kauditd
   33 ?           00:00:00 khungtaskd
   34 ?           00:00:00 oom_reaper
   35 ?           00:00:00 kworker/u6:1-events_unbound
   36 ?           00:00:00 kworker/R-write
   37 ?           00:00:00 kcompactd0
   38 ?           00:00:00 ksmd
   40 ?           00:00:00 khugepaged
   41 ?           00:00:00 kworker/R-kinte
   42 ?           00:00:00 kworker/R-kbloc
   43 ?           00:00:00 kworker/R-blkcg
   44 ?           00:00:00 irq/9-acpi
   45 ?           00:00:00 kworker/R-tpm_d
   46 ?           00:00:00 kworker/R-ata_s
   47 ?           00:00:00 kworker/R-md
   48 ?           00:00:00 kworker/R-md_bi
   49 ?           00:00:00 kworker/R-edac-
   50 ?           00:00:00 kworker/R-devfr
   51 ?           00:00:00 watchdogd
   53 ?           00:00:00 kswapd0
   54 ?           00:00:00 ecryptfs-kthread
   55 ?           00:00:00 kworker/R-kthro
   56 ?           00:00:00 kworker/R-acpi_
   57 ?           00:00:00 scsi_eh_0
   58 ?           00:00:00 kworker/R-scsi

```

59 ?	00:00:00	scsi_eh_1
60 ?	00:00:00	kworker/R-scsi_
62 ?	00:00:00	kworker/u5:3-flush-8:0
63 ?	00:00:00	kworker/R-mld
64 ?	00:00:00	kworker/0:1H-kblockd
65 ?	00:00:00	kworker/R-ipv6_
67 ?	00:00:00	kworker/u5:4-flush-8:0
73 ?	00:00:00	kworker/R-kstrp
75 ?	00:00:00	kworker/u7:0
76 ?	00:00:00	kworker/u8:0
77 ?	00:00:00	kworker/u9:0
82 ?	00:00:00	kworker/R-crypt
93 ?	00:00:00	kworker/R-charg
137 ?	00:00:00	kworker/R-mpt_p
138 ?	00:00:00	kworker/R-mpt/0
156 ?	00:00:00	scsi_eh_2
157 ?	00:00:00	kworker/R-scsi_
191 ?	00:00:00	kworker/R-raid5
214 ?	00:00:00	kworker/0:2H-kblockd
229 ?	00:00:00	jbd2/sda1-8
230 ?	00:00:00	kworker/R-ext4-
297 ?	00:00:00	psimon
302 ?	00:00:00	systemd-journal
334 ?	00:00:00	kworker/R-kmpat
335 ?	00:00:00	kworker/R-kmpat
338 ?	00:00:00	kworker/0:3-cgroup_destroy
344 ?	00:00:00	kworker/u6:2-flush-8:0
357 ?	00:00:00	multipathd
366 ?	00:00:00	systemd-udevd
370 ?	00:00:00	kworker/1:4-cgwb_release
379 ?	00:00:00	psimon
454 ?	00:00:00	jbd2/sda16-8
455 ?	00:00:00	kworker/R-ext4-
460 ?	00:00:00	irq/18-vmwgfx
461 ?	00:00:00	kworker/R-ttm
495 ?	00:00:00	systemd-network
516 ?	00:00:00	systemd-resolve
524 ?	00:00:00	systemd-timesyn
644 ?	00:00:00	dbus-daemon
650 ?	00:00:00	polkitd
663 ?	00:00:00	systemd-logind
665 ?	00:00:00	udisksd
681 ?	00:00:00	kworker/u4:2-ext4-rsv-conversion
698 ?	00:00:00	kworker/1:2H-kblockd
710 ?	00:00:00	unattended-upgr
740 ?	00:00:00	ModemManager
749 ?	00:00:00	rsyslogd
783 ?	00:00:00	cron
821 tty1	00:00:00	agetty
879 ?	00:00:00	kworker/u6:4-flush-8:0
909 ?	00:00:00	sshd
911 ?	00:00:00	sshd
913 ?	00:00:01	sshd
914 pts/0	00:00:00	bash

1034 ?	00:00:00	kworker/u5:0-events_unbound
1073 ?	00:00:00	kworker/0:1
1084 pts/0	00:00:00	mainprocess
1085 pts/0	00:00:00	mainprocess
1086 pts/0	00:00:00	mainprocess
1087 pts/0	00:00:00	subprocess2
1088 pts/0	00:00:00	subprocess2
1089 pts/0	00:00:00	subprocess1
1090 pts/0	00:00:00	subprocess2
1091 pts/0	00:00:00	subprocess1
1092 pts/0	00:00:00	ps

Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

Command
<code>ps aux</code>

Output

```
syahmi@secr2043:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.2  1.3 21996 13056 ?        Ss   15:08   0:01 /sbin/init
root         2  0.0  0.0      0     0 ?        S    15:08   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    15:08   0:00 [pool_workqueue_rele
root         4  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-rcu_g]
root         5  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-rcu_p]
root         6  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-slab_]
root         7  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-netns]
root         8  0.2  0.0      0     0 ?        I    15:08   0:02 [kworker/0:0-mm_perc
root        11  0.0  0.0      0     0 ?        I    15:08   0:00 [kworker/u4:0-floppy
root        12  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-mm_pe]
root        13  0.0  0.0      0     0 ?        I    15:08   0:00 [rcu_tasks_kthread]
root        14  0.0  0.0      0     0 ?        I    15:08   0:00 [rcu_tasks_rude_kthr
root        15  0.0  0.0      0     0 ?        I    15:08   0:00 [rcu_tasks_trace_kth
root        16  0.0  0.0      0     0 ?        S    15:08   0:00 [ksoftirqd/0]
root        17  0.0  0.0      0     0 ?        I    15:08   0:00 [rcu_preempt]
root        18  0.0  0.0      0     0 ?        S    15:08   0:00 [migration/0]
root        19  0.0  0.0      0     0 ?        S    15:08   0:00 [idle_inject/0]
root        20  0.0  0.0      0     0 ?        S    15:08   0:00 [cpuhp/0]
root        21  0.0  0.0      0     0 ?        S    15:08   0:00 [cpuhp/1]
root        22  0.0  0.0      0     0 ?        S    15:08   0:00 [idle_inject/1]
root        23  0.0  0.0      0     0 ?        S    15:08   0:00 [migration/1]
root        24  0.0  0.0      0     0 ?        S    15:08   0:00 [ksoftirqd/1]
root        25  0.4  0.0      0     0 ?        I    15:08   0:03 [kworker/1:0-mm_perc
root        26  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/1:0H-events
root        29  0.0  0.0      0     0 ?        S    15:08   0:00 [kdevtmpfs]
root        30  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-inet_]
root        32  0.0  0.0      0     0 ?        S    15:08   0:00 [kauditd]
root        33  0.0  0.0      0     0 ?        S    15:08   0:00 [khungtaskd]
root        34  0.0  0.0      0     0 ?        S    15:08   0:00 [oom_reaper]
root        35  0.0  0.0      0     0 ?        I    15:08   0:00 [kworker/u6:1-events
root        36  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-write]
root        37  0.0  0.0      0     0 ?        S    15:08   0:00 [kcompactd0]
root        38  0.0  0.0      0     0 ?        SN   15:08   0:00 [ksmd]
root        40  0.0  0.0      0     0 ?        SN   15:08   0:00 [khugepaged]
root        41  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-kinte]
root        42  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-kbloc]
root        43  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-blkcg]
root        44  0.0  0.0      0     0 ?        S    15:08   0:00 [irq/9-acpi]
root        45  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-tpm_d]
root        46  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-ata_s]
root        47  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-md]
root        48  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-md_bi]
root        49  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-edac_]
root        50  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-devfr]
root        51  0.0  0.0      0     0 ?        S    15:08   0:00 [watchdogd]
root        53  0.0  0.0      0     0 ?        S    15:08   0:00 [kswapd0]
root        54  0.0  0.0      0     0 ?        S    15:08   0:00 [ecryptfs-kthread]
root        55  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-kthro]
root        56  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-acpi_]
root        57  0.0  0.0      0     0 ?        S    15:08   0:00 [scsi_eh_0]
root        58  0.0  0.0      0     0 ?        I<   15:08   0:00 [kworker/R-scsi_]

```

root	59	0.0	0.0	0	0	?	S	15:08	0:00	[scsi_eh_1]
root	60	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/R-scsi_]
root	62	0.0	0.0	0	0	?	I	15:08	0:00	[kworker/u5:3-flush-
root	63	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/R-mld]
root	64	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/0:1H-kblock
root	65	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/R-ipv6_]
root	67	0.0	0.0	0	0	?	I	15:08	0:00	[kworker/u5:4-events
root	73	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/R-kstrp]
root	75	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/u7:0]
root	76	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/u8:0]
root	77	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/u9:0]
root	82	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/R-crypt]
root	93	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/R-charg]
root	137	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/R-mpt_p]
root	138	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/R-mpt/0]
root	156	0.0	0.0	0	0	?	S	15:08	0:00	[scsi_eh_2]
root	157	0.0	0.0	0	0	?	I<	15:08	0:00	[kworker/R-scsi_]
root	191	0.0	0.0	0	0	?	I<	15:09	0:00	[kworker/R-raid5]
root	214	0.0	0.0	0	0	?	I<	15:09	0:00	[kworker/0:2H-kblock
root	229	0.0	0.0	0	0	?	S	15:09	0:00	[jbd2/sda1-8]
root	230	0.0	0.0	0	0	?	I<	15:09	0:00	[kworker/R-ext4-]
root	297	0.0	0.0	0	0	?	S	15:09	0:00	[psimon]
root	302	0.0	1.2	34036	11912	?	S<S	15:09	0:00	/usr/lib/systemd/sys
root	334	0.0	0.0	0	0	?	I<	15:09	0:00	[kworker/R-kmpat]
root	335	0.0	0.0	0	0	?	I<	15:09	0:00	[kworker/R-kmpat]
root	338	0.0	0.0	0	0	?	I	15:09	0:00	[kworker/0:3-cgroup_
root	344	0.0	0.0	0	0	?	I	15:09	0:00	[kworker/u6:2-flush-
root	357	0.0	2.7	354524	27264	?	SLsL	15:09	0:00	/sbin/multipathd -d
root	366	0.0	0.7	25952	7552	?	Ss	15:09	0:00	/usr/lib/systemd/sys
root	370	0.0	0.0	0	0	?	I	15:09	0:00	[kworker/1:4-cgwb_re
root	379	0.0	0.0	0	0	?	S	15:09	0:00	[psimon]
root	454	0.0	0.0	0	0	?	S	15:09	0:00	[jbd2/sda16-8]
root	455	0.0	0.0	0	0	?	I<	15:09	0:00	[kworker/R-ext4-]
root	460	0.0	0.0	0	0	?	S	15:09	0:00	[irq/18-vmwgfx]
root	461	0.0	0.0	0	0	?	I<	15:09	0:00	[kworker/R-ttm]
systemd+	495	0.0	0.9	18996	9472	?	Ss	15:09	0:00	/usr/lib/systemd/sys
systemd+	516	0.0	1.3	21576	12800	?	Ss	15:09	0:00	/usr/lib/systemd/sys
systemd+	524	0.0	0.7	91020	7808	?	SsL	15:09	0:00	/usr/lib/systemd/sys
message+	644	0.0	0.5	9768	5376	?	Ss	15:09	0:00	@dbus-daemon --syste
polkitd	650	0.0	0.7	308160	7808	?	SsL	15:09	0:00	/usr/lib/polkit-1/po
root	663	0.0	0.8	17620	7936	?	Ss	15:09	0:00	/usr/lib/systemd/sys
root	665	0.0	1.3	468956	13568	?	SsL	15:09	0:00	/usr/libexec/udisks2
root	681	0.0	0.0	0	0	?	I	15:09	0:00	[kworker/u4:2-ext4-r
root	698	0.0	0.0	0	0	?	I<	15:09	0:00	[kworker/1:2H-kblock
root	710	0.0	2.3	109996	22784	?	SsL	15:09	0:00	/usr/bin/python3 /us
root	740	0.0	1.3	392092	12800	?	SsL	15:09	0:00	/usr/sbin/ModemManag
syslog	749	0.0	0.5	222508	5760	?	SsL	15:09	0:00	/usr/sbin/rsyslogd -
root	783	0.0	0.2	7224	2688	?	Ss	15:09	0:00	/usr/sbin/cron -f -P
root	821	0.0	0.1	6104	1920	tty1	Ss+	15:09	0:00	/sbin/agetty -o -p -
root	879	0.0	0.0	0	0	?	I	15:09	0:00	[kworker/u6:4-flush-
root	909	0.0	0.8	12016	7936	?	Ss	15:11	0:00	sshd: /usr/sbin/sshd
root	911	0.0	0.5	12148	5140	?	Ss	15:11	0:00	sshd: syahmi [priv]
syahmi	913	0.1	0.7	13760	7172	?	S	15:11	0:01	sshd: syahmi@pts/0
syahmi	914	0.0	0.5	5644	4992	pts/0	Ss	15:11	0:00	-bash
root	1034	0.0	0.0	0	0	?	I	15:18	0:00	[kworker/u5:0-events
root	1073	0.0	0.0	0	0	?	I	15:19	0:00	[kworker/0:1]
syahmi	1084	0.0	0.1	2680	1536	pts/0	S	15:20	0:00	./mainprocess
syahmi	1085	0.0	0.0	2680	128	pts/0	S	15:20	0:00	./mainprocess
syahmi	1086	0.0	0.0	2680	128	pts/0	S	15:20	0:00	./mainprocess
syahmi	1087	0.0	0.1	2680	1536	pts/0	S	15:20	0:00	./subprocess2
syahmi	1088	0.0	0.1	2680	1408	pts/0	S	15:20	0:00	./subprocess2
syahmi	1089	0.0	0.1	2680	1536	pts/0	S	15:20	0:00	./subprocess1
syahmi	1090	0.0	0.1	2680	1536	pts/0	S	15:20	0:00	./subprocess2
syahmi	1091	0.0	0.1	2680	1536	pts/0	S	15:20	0:00	./subprocess1
syahmi	1093	0.0	0.4	7892	4096	pts/0	R+	15:23	0:00	ps aux

Question 3

Use the ps command with some tools to only list processes named "subprocess" and show some info about them.

Command
ps aux grep subprocess
Output

```
syahmi@secr2043:~$ ps aux | grep subprocess
syahmi    1087  0.0  0.1   2680  1536 pts/0    S      15:20   0:00  ./subprocess2
syahmi    1088  0.0  0.1   2680  1408 pts/0    S      15:20   0:00  ./subprocess2
syahmi    1089  0.0  0.1   2680  1536 pts/0    S      15:20   0:00  ./subprocess1
syahmi    1090  0.0  0.1   2680  1536 pts/0    S      15:20   0:00  ./subprocess2
syahmi    1091  0.0  0.1   2680  1536 pts/0    S      15:20   0:00  ./subprocess1
syahmi    1103  0.0  0.1   3528  1792 pts/0    S+     15:25   0:00  grep --color=auto su
bprocess
```

Question 4

Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)
- Command (`cmd`)

Command
<code>ps -eo pid,user,pcpu,pmem,cmd</code>
Output

```

syahmi@secr2043:~$ ps -eo pid,user,pcpu,pmem,cmd
  PID USER      %CPU %MEM    CMD
    1 root         0.1   1.3  /sbin/init
    2 root         0.0   0.0  [kthreadd]
    3 root         0.0   0.0  [pool_workqueue_release]
    4 root         0.0   0.0  [kworker/R-rcu_g]
    5 root         0.0   0.0  [kworker/R-rcu_p]
    6 root         0.0   0.0  [kworker/R-slub_]
    7 root         0.0   0.0  [kworker/R-netns]
    8 root         0.3   0.0  [kworker/0:0-mpt_poll_0]
   11 root         0.0   0.0  [kworker/u4:0-floppy]
   12 root         0.0   0.0  [kworker/R-mm_pe]
   13 root         0.0   0.0  [rcu_tasks_kthread]
   14 root         0.0   0.0  [rcu_tasks_rude_kthread]
   15 root         0.0   0.0  [rcu_tasks_trace_kthread]
   16 root         0.0   0.0  [ksoftirqd/0]
   17 root         0.0   0.0  [rcu_preempt]
   18 root         0.0   0.0  [migration/0]
   19 root         0.0   0.0  [idle_inject/0]
   20 root         0.0   0.0  [cpuhp/0]
   21 root         0.0   0.0  [cpuhp/1]
   22 root         0.0   0.0  [idle_inject/1]
   23 root         0.0   0.0  [migration/1]
   24 root         0.0   0.0  [ksoftirqd/1]
   25 root         0.3   0.0  [kworker/1:0-events]
   26 root         0.0   0.0  [kworker/1:0H-events_highpri]
   29 root         0.0   0.0  [kdevtmpfs]
   30 root         0.0   0.0  [kworker/R-inet_]
   32 root         0.0   0.0  [kauditd]
   33 root         0.0   0.0  [khungtaskd]
   34 root         0.0   0.0  [oom_reaper]
   35 root         0.0   0.0  [kworker/u6:1-events_unbound]
   36 root         0.0   0.0  [kworker/R-write]
   37 root         0.0   0.0  [kcompactd0]
   38 root         0.0   0.0  [ksmd]
   40 root         0.0   0.0  [khugepaged]
   41 root         0.0   0.0  [kworker/R-kinte]
   42 root         0.0   0.0  [kworker/R-kbloc]
   43 root         0.0   0.0  [kworker/R-blkcg]
   44 root         0.0   0.0  [irq/9-acpi]
   45 root         0.0   0.0  [kworker/R-tpm_d]
   46 root         0.0   0.0  [kworker/R-ata_s]
   47 root         0.0   0.0  [kworker/R-md]
   48 root         0.0   0.0  [kworker/R-md_bi]
   49 root         0.0   0.0  [kworker/R-edac-]
   50 root         0.0   0.0  [kworker/R-devfr]
   51 root         0.0   0.0  [watchdogd]
   53 root         0.0   0.0  [kswapd0]
   54 root         0.0   0.0  [ecryptfs-kthread]
   55 root         0.0   0.0  [kworker/R-kthro]

```



```

56 root      0.0  0.0 [kworker/R-acpi_]
57 root      0.0  0.0 [scsi_eh_0]
58 root      0.0  0.0 [kworker/R-scsi_]
59 root      0.0  0.0 [scsi_eh_1]
60 root      0.0  0.0 [kworker/R-scsi_]
62 root      0.0  0.0 [kworker/u5:3-flush-8:0]
63 root      0.0  0.0 [kworker/R-mld]
64 root      0.0  0.0 [kworker/0:1H-kblockd]
65 root      0.0  0.0 [kworker/R-ipv6_]
67 root      0.0  0.0 [kworker/u5:4-events_power_efficient]
73 root      0.0  0.0 [kworker/R-kstrp]
75 root      0.0  0.0 [kworker/u7:0]
76 root      0.0  0.0 [kworker/u8:0]
77 root      0.0  0.0 [kworker/u9:0]
82 root      0.0  0.0 [kworker/R-crypt]
93 root      0.0  0.0 [kworker/R-charge]
137 root     0.0  0.0 [kworker/R-mpt_p]
138 root     0.0  0.0 [kworker/R-mpt/0]
156 root     0.0  0.0 [scsi_eh_2]
157 root     0.0  0.0 [kworker/R-scsi_]
191 root     0.0  0.0 [kworker/R-raid5]
214 root     0.0  0.0 [kworker/0:2H-kblockd]
229 root     0.0  0.0 [jbd2/sda1-8]
230 root     0.0  0.0 [kworker/R-ext4-]
297 root     0.0  0.0 [psimon]
302 root     0.0  1.2 /usr/lib/systemd/systemd-journald
334 root     0.0  0.0 [kworker/R-kmpat]
335 root     0.0  0.0 [kworker/R-kmpat]
344 root     0.0  0.0 [kworker/u6:2-flush-8:0]
357 root     0.0  2.7 /sbin/multipathd -d -s
366 root     0.0  0.7 /usr/lib/systemd/systemd-udev
370 root     0.0  0.0 [kworker/1:4-cgwb_release]
379 root     0.0  0.0 [psimon]
454 root     0.0  0.0 [jbd2/sda16-8]
455 root     0.0  0.0 [kworker/R-ext4-]
460 root     0.0  0.0 [irq/18-vmwgfx]
461 root     0.0  0.0 [kworker/R-ttm]
495 systemd+ 0.0  0.9 /usr/lib/systemd/systemd-networkd
516 systemd+ 0.0  1.3 /usr/lib/systemd/systemd-resolved
524 systemd+ 0.0  0.7 /usr/lib/systemd/systemd-timesyncd
644 message+ 0.0  0.5 @dbus-daemon --system --address=systemd: --nofork --nopidfil
650 polkitd   0.0  0.7 /usr/lib/polkit-1/polkitd --no-debug
663 root     0.0  0.8 /usr/lib/systemd/systemd-logind
665 root     0.0  1.3 /usr/libexec/udisks2/udisksd
681 root     0.0  0.0 [kworker/u4:2-ext4-rsv-conversion]
698 root     0.0  0.0 [kworker/1:2H-kblockd]
710 root     0.0  2.3 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-u
740 root     0.0  1.3 /usr/sbin/ModemManager
749 syslog   0.0  0.5 /usr/sbin/rsyslogd -n -iNONE
783 root     0.0  0.2 /usr/sbin/cron -f -P
821 root     0.0  0.1 /sbin/agetty -o -p -- \u --noclear - linux
879 root     0.0  0.0 [kworker/u6:4-events_power_efficient]
909 root     0.0  0.8 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
911 root     0.0  0.5 sshd: syahmi [priv]

913 syahmi   0.1  0.7 sshd: syahmi@pts/0
914 syahmi   0.0  0.5 -bash
1034 root    0.0  0.0 [kworker/u5:0-events_unbound]
1073 root    0.0  0.0 [kworker/0:1-cgroup_destroy]
1084 syahmi   0.0  0.1 ./mainprocess
1085 syahmi   0.0  0.0 ./mainprocess
1086 syahmi   0.0  0.0 ./mainprocess
1087 syahmi   0.0  0.1 ./subprocess2
1088 syahmi   0.0  0.1 ./subprocess2
1089 syahmi   0.0  0.1 ./subprocess1
1090 syahmi   0.0  0.1 ./subprocess2
1091 syahmi   0.0  0.1 ./subprocess1
1101 root     0.0  0.0 [kworker/u6:0]
1104 syahmi   0.0  0.4 ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem

```

Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (pmem)?

Command

```
ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
```

Output

```
syahmi@seacr2043:~$ ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
PID USER      %CPU %MEM    CMD
357 root        0.0  2.7  /sbin/multipathd -d -s
710 root        0.0  2.3  /usr/bin/python3 /usr/share/unattended-upgrades/unattended-u
665 root        0.0  1.3  /usr/libexec/udisks2/udisksd
1 root         0.1  1.3  /sbin/init
516 systemd+  0.0  1.3  /usr/lib/systemd/systemd-resolved
740 root        0.0  1.3  /usr/sbin/ModemManager
302 root        0.0  1.2  /usr/lib/systemd/systemd-journald
495 systemd+  0.0  0.9  /usr/lib/systemd/systemd-networkd
663 root        0.0  0.8  /usr/lib/systemd/systemd-logind
909 root        0.0  0.8  sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
524 systemd+  0.0  0.7  /usr/lib/systemd/systemd-timesyncd
650 polkitd    0.0  0.7  /usr/lib/polkit-1/polkitd --no-debug
366 root        0.0  0.7  /usr/lib/systemd/systemd-udev
913 syahmi      0.1  0.7  sshd: syahmi@pts/0
749 syslog     0.0  0.5  /usr/sbin/rsyslogd -n -iNONE
644 message+   0.0  0.5  @dbus-daemon --system --address=systemd: --nofork --nopidfil
911 root        0.0  0.5  sshd: syahmi [priv]
914 syahmi     0.0  0.5  -bash
1105 syahmi    100  0.4  ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
783 root        0.0  0.2  /usr/sbin/cron -f -P
821 root        0.0  0.1  /sbin/agetty -o -p -- \u --noclear - linux
1084 syahmi    0.0  0.1  ./mainprocess
1087 syahmi    0.0  0.1  ./subprocess2
1089 syahmi    0.0  0.1  ./subprocess1
1090 syahmi    0.0  0.1  ./subprocess2
1091 syahmi    0.0  0.1  ./subprocess1
1088 syahmi    0.0  0.1  ./subprocess2
1085 syahmi    0.0  0.0  ./mainprocess
1086 syahmi    0.0  0.0  ./mainprocess
2 root        0.0  0.0  [kthreadd]
3 root        0.0  0.0  [pool_workqueue_release]
4 root        0.0  0.0  [kworker/R-rcu_g]
5 root        0.0  0.0  [kworker/R-rcu_p]
6 root        0.0  0.0  [kworker/R-slub_]
7 root        0.0  0.0  [kworker/R-netns]
8 root        0.3  0.0  [kworker/0:0-ata_sff]
11 root       0.0  0.0  [kworker/u4:0-floppy]
12 root       0.0  0.0  [kworker/R-mm_pe]
13 root       0.0  0.0  [rcu_tasks_kthread]
14 root       0.0  0.0  [rcu_tasks_rude_kthread]
15 root       0.0  0.0  [rcu_tasks_trace_kthread]
16 root       0.0  0.0  [ksoftirqd/0]
17 root       0.0  0.0  [rcu_preempt]
18 root       0.0  0.0  [migration/0]
19 root       0.0  0.0  [idle_inject/0]
20 root       0.0  0.0  [cpuhp/0]
21 root       0.0  0.0  [cpuhp/1]
22 root       0.0  0.0  [idle_inject/1]
23 root       0.0  0.0  [migration/1]
24 root       0.0  0.0  [ksoftirqd/1]
25 root       0.3  0.0  [kworker/1:0-events]
```

```

26 root      0.0  0.0 [kworker/1:0H-events_highpri]
29 root      0.0  0.0 [kdevtmpfs]
30 root      0.0  0.0 [kworker/R-inet_]
32 root      0.0  0.0 [kauditd]
33 root      0.0  0.0 [khungtaskd]
34 root      0.0  0.0 [oom_reaper]
35 root      0.0  0.0 [kworker/u6:1-events_unbound]
36 root      0.0  0.0 [kworker/R-write]
37 root      0.0  0.0 [kcompactd0]
38 root      0.0  0.0 [ksmd]
40 root      0.0  0.0 [khugepaged]
41 root      0.0  0.0 [kworker/R-kinte]
42 root      0.0  0.0 [kworker/R-kbloc]
43 root      0.0  0.0 [kworker/R-blkcg]
44 root      0.0  0.0 [irq/9-acpi]
45 root      0.0  0.0 [kworker/R-tpm_d]
46 root      0.0  0.0 [kworker/R-ata_s]
47 root      0.0  0.0 [kworker/R-md]
48 root      0.0  0.0 [kworker/R-md_bi]
49 root      0.0  0.0 [kworker/R-edac-]
50 root      0.0  0.0 [kworker/R-devfr]
51 root      0.0  0.0 [watchdogd]
53 root      0.0  0.0 [kswapd0]
54 root      0.0  0.0 [ecryptfs-kthread]
55 root      0.0  0.0 [kworker/R-kthro]
56 root      0.0  0.0 [kworker/R-acpi_]
57 root      0.0  0.0 [scsi_eh_0]
58 root      0.0  0.0 [kworker/R-scsi_]
59 root      0.0  0.0 [scsi_eh_1]
60 root      0.0  0.0 [kworker/R-scsi_]
62 root      0.0  0.0 [kworker/u5:3-flush-8:0]
63 root      0.0  0.0 [kworker/R-mld]
64 root      0.0  0.0 [kworker/0:1H-kblockd]
65 root      0.0  0.0 [kworker/R-ipv6_]
67 root      0.0  0.0 [kworker/u5:4-events_power_efficient]
73 root      0.0  0.0 [kworker/R-kstrp]
75 root      0.0  0.0 [kworker/u7:0]
76 root      0.0  0.0 [kworker/u8:0]
77 root      0.0  0.0 [kworker/u9:0]
82 root      0.0  0.0 [kworker/R-crypt]
93 root      0.0  0.0 [kworker/R-charg]
137 root     0.0  0.0 [kworker/R-mpt_p]
138 root     0.0  0.0 [kworker/R-mpt/0]
156 root     0.0  0.0 [scsi_eh_2]
157 root     0.0  0.0 [kworker/R-scsi_]
191 root     0.0  0.0 [kworker/R-raid5]
214 root     0.0  0.0 [kworker/0:2H-kblockd]
229 root     0.0  0.0 [jbd2/sda1-8]
230 root     0.0  0.0 [kworker/R-ext4-]
297 root     0.0  0.0 [psimon]
334 root     0.0  0.0 [kworker/R-kmpat]
335 root     0.0  0.0 [kworker/R-kmpat]
344 root     0.0  0.0 [kworker/u6:2-flush-8:0]
370 root     0.0  0.0 [kworker/1:4-cqwb_release]

379 root     0.0  0.0 [psimon]
454 root     0.0  0.0 [jbd2/sda16-8]
455 root     0.0  0.0 [kworker/R-ext4-]
460 root     0.0  0.0 [irq/18-vmwgfx]
461 root     0.0  0.0 [kworker/R-ttm]
681 root     0.0  0.0 [kworker/u4:2-ext4-rsv-conversion]
698 root     0.0  0.0 [kworker/1:2H-kblockd]
879 root     0.0  0.0 [kworker/u6:4-events_power_efficient]
1034 root    0.0  0.0 [kworker/u5:0-events_unbound]
1073 root    0.0  0.0 [kworker/0:1-cgroup_destroy]
1101 root    0.0  0.0 [kworker/u6:0]

```

Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

Command

ps -ejH | grep -E 'mainprocess|subprocess1|subprocess2'

Output

```
syahmi@secr2043:~$ ps -ejH | grep -E 'mainprocess|subprocess1|subprocess2'
```

1084	1084	914	pts/0	00:00:00	mainprocess
1085	1084	914	pts/0	00:00:00	mainprocess
1089	1084	914	pts/0	00:00:00	subprocess1
1091	1084	914	pts/0	00:00:00	subprocess1
1086	1084	914	pts/0	00:00:00	mainprocess
1087	1084	914	pts/0	00:00:00	subprocess2
1088	1084	914	pts/0	00:00:00	subprocess2
1090	1084	914	pts/0	00:00:00	subprocess2

Question 7

Use `pstree` command with option that show the number of threads to each process.

Command	
pstree -pT	
Output	
<pre>syahmi@secr2043:~\$ pstree -pT</pre> <pre> systemd(1)-+-ModemManager(740) --agetty(821) --cron(783) --dbus-daemon(644) --multipathd(357) --polkitd(650) --rsyslogd(749) --sshd(909)---sshd(911)---sshd(913)---bash(914)-+-mainprocess(1084)-+-mainp+ '-mainp+ --pstree(1109) --systemd-journal(302) --systemd-logind(663) --systemd-network(495) --systemd-resolve(516) --systemd-timesyn(524) --systemd-udev(366) --udisksd(665) `--unattended-upgr(710) </pre>	

Question 8

Use `renice` command to change priority level of one of process “subprocess1”.

Command
<pre>ps aux grep subprocess1 sudo renice -n 10 -p 895</pre>
Output
<pre>syahmi@secr2043:~\$ ps aux grep subprocess1 syahmi 1089 0.0 0.1 2680 1536 pts/0 S 15:20 0:00 ./subprocess1 syahmi 1091 0.0 0.1 2680 1536 pts/0 S 15:20 0:00 ./subprocess1 syahmi 1115 0.0 0.1 3528 1792 pts/0 S+ 15:30 0:00 grep --color=auto su bprocess1 syahmi@secr2043:~\$ sudo renice -n 10 -p 1089 1089 (process ID) old priority 0, new priority 10</pre>

Question 9

Terminate all running processes with the name “mainprocess”.

Command
<code>killall mainprocess</code>
Output
<pre>syahmi@secr2043:~\$ killall mainprocess Main process (ID: 1085) received signal: 15. Terminating... Main process (ID: 1084) received signal: 15. Terminating... Main process (ID: 1086) received signal: 15. Terminating...</pre>

Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        // Fork failed
        perror("Fork failed");
        return 1;
    } else if (pid == 0) {
        // Child process
        printf("Hello, world! This is the child process.\n");
    } else {
        // Parent process
        wait(NULL); // Wait for child to finish
        printf("This is the parent process.\n");
    }

    return 0;
}
```

Output:

```
syahmi@secr2043:~$ nano fork_example.c
syahmi@secr2043:~$ gcc fork_example.c -o fork_example
syahmi@secr2043:~$ ./fork_example
Hello, world! This is the child process.
This is the parent process.
```